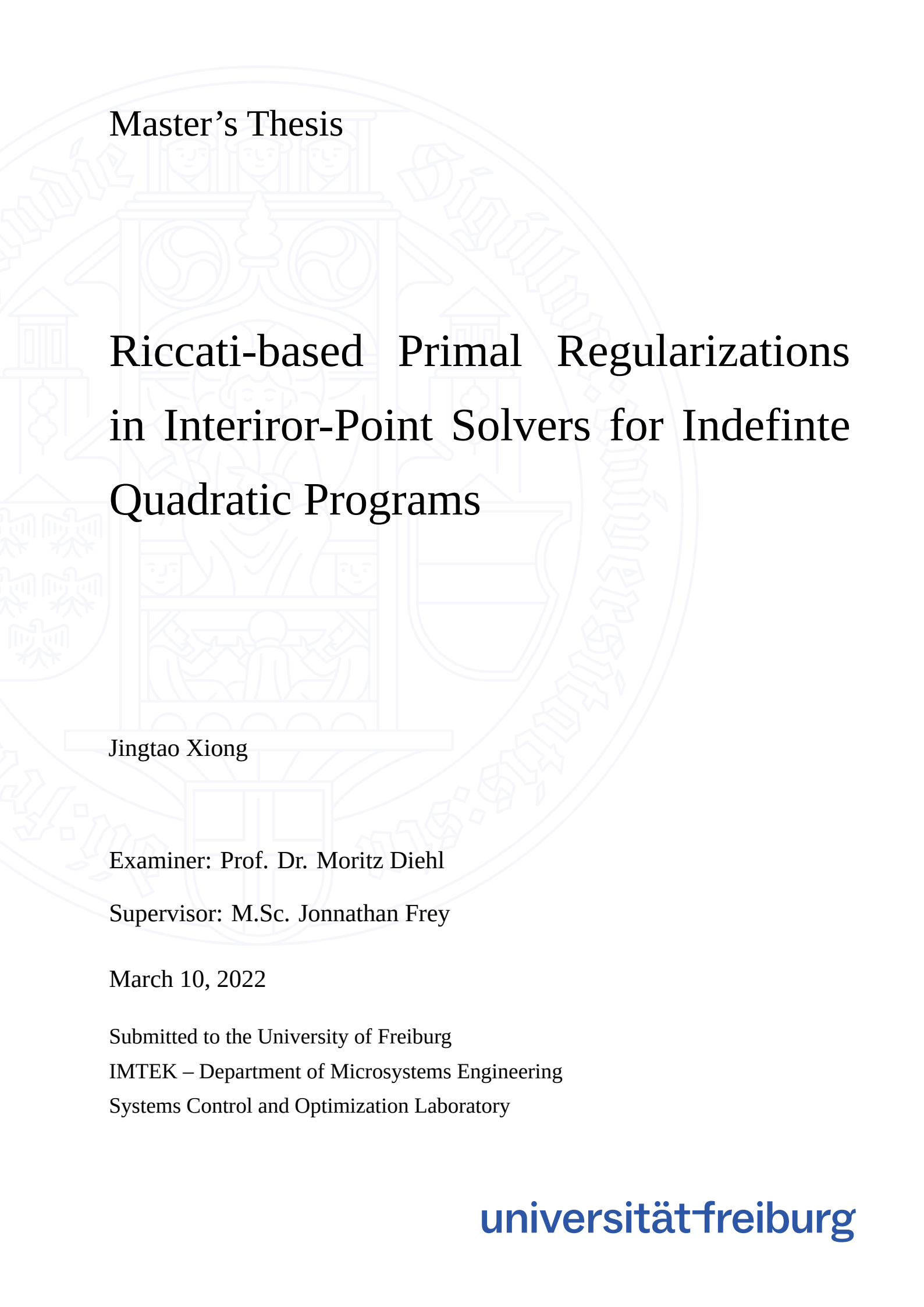




Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Chapter 1

Primal Regularization for Indefinite Quadratic Programs

In this chapter, we turn our attention to the QPs with indefinite Hessian matrix, namely indefinite QPs. The chapter starts with a brief classification of the Hessian regularization in 1.1, then we introduce the concept of Riccati-based primal regularization in 1.2, the Riccati-based primal regularization method is consecutively introduced based on an analysis of failing on minimum perturbation in 1.3. We conclude the chapter with a proof of the asymptotic exactness of the proposed Riccati-based primal regularization method.

1.1 Primal regularization

In indefinite QPs, the regularization needed to be applied to the Hessian matrix to ensure a descent search direction in the interior point method. The resulting regularized reduced KKT system can be generally presented as:

$$\begin{bmatrix} \tilde{H} + E & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}_{\text{stat}}^i \\ r_{\text{eq}}^i \end{bmatrix}, \quad (1.1)$$

where E is the adjustment matrix for ensuring a positive definite Hessian. This process is commonly referred to as primal regularization and this section will briefly introduce the two types of primal regularization, i.e. the regularization on full Hessian $\tilde{H}$ and the regularization on reduced Hessian $\tilde{M}$.

1.1.1 Regularization Strategies

We would like to start with the numerical techniques used by both full Hessian regularization and reduced Hessian regularization by regarding the calculation of adjustment matrix E .

There are several strategies for constructing the adjustment matrix E to ensure a positive definite Hessian, commonly, it can be computed from

- Eigenvalue modification (EM): modifies the negative eigenvalues of $\tilde{H}$ [1].
- Modified Cholesky factorization (MC): modifies the inertia of $\tilde{H}$ [2].
- Levenberg-Marquardt (LM): adds a scaled identity matrix to $\tilde{H}$ [3].

We will mainly apply four commonly used strategies for regularization in this work, two methods for EM (PROJECT, MIRROR) [4], one method for MC (GMW) [5], and one method for LM (Inertia Correction) [6].

Assuming a matrix A can be eigenvalue decomposed as $A = VD_EV^\top$ and Cholesky factorized as $A = LD_IL^\top$, where V is an orthogonal matrix of eigenvectors, L is a unit lower triangular matrix, D_E is a diagonal matrix of real eigenvalues, and D_I is a diagonal matrix whose signs give the inertia of A

The four methods can be summarized as follows:

- **PROJECT** (Minimal eigenvalue modification):

$$\text{project}(A, \epsilon) := V \max(\epsilon I, D_E) V^\top.$$

Where the max operator is applied component-wise. This approach clips any negative or near-zero eigenvalues to a small positive constant $\epsilon > 0$, which keeps a minimal adjustment to the original matrix for positive definiteness, but may lead to an increase in the condition number if the original eigenvalues were not close to zero.

- **MIRROR** (Absolute eigenvalue modification):

$$\text{mirror}(A, \epsilon) := V \max(\epsilon I, |D_E|) V^\top.$$

This method reflects negative eigenvalues to their positive counterparts. It preserves the magnitude of the curvature in all directions, which often provides better scaling, but may result in a large modification to the original matrix.

- **GMW** (Modified Cholesky factorization): Instead of a spectral decomposition, GMW modifies the diagonal pivots on the fly during the $LDL^\top$

factorization. At the j -th step, given the trial pivot c_{jj} and the largest current off-diagonal $\theta_j = \max_{i>j} |c_{ij}|$, the pivot is bounded from below by

$$d_j^I = \max\left(\epsilon, |c_{jj}|, \frac{\theta_j^2}{\beta^2}\right), \quad \beta^2 = \max\left(\gamma, \frac{\xi}{\sqrt{n^2-1}}, \epsilon_M\right),$$

where $\gamma = \max_i |a_{ii}|$, $\xi = \max_{i \neq j} |a_{ij}|$, and ϵ_M is the machine precision. The term θ_j^2/β^2 bounds the entries of L and thus the condition number of the factor, while the implied correction is $e_j = d_j^I - c_{jj} \geq 0$, giving $A + E = LD_I L^\top$.

This avoids the cost of an eigendecomposition and still guarantees positive definiteness, but the resulting E is generally not minimal and depends on the variable ordering, so it tends to over-modify A compared with the eigenvalue-based methods.

- **Inertia Correction** (Levenberg-Marquardt):

$$\text{ic}(A, \Delta) := A + \Delta I, \quad \Delta \geq 0.$$

A uniform multiple of the identity is added, so every eigenvalue is shifted by the same amount, $\lambda_k(A + \Delta I) = \lambda_k(A) + \Delta$. In practice $\lambda_{\min}$ is not computed directly, and Δ is found by repeatedly attempting the $\text{LDL}^\top$ factorization and increasing Δ until the inertia is correct, i.e. until every diagonal entry of D_I is positive. The shift leaves the eigenvectors unchanged and is very cheap and robust, however, because it acts equally in all directions, it also inflates the already-positive eigenvalues and can degrade the scaling when only a few directions are indefinite.

1.1.2 Full Hessian regularization

The full hessian regularization aims at the positive definiteness of the Hessian matrix $\tilde{H}$. Commonly, in the optimal control structure, the full Hessian is regularized stage-wisely by operating on the block Hessian matrix $\tilde{H}_k$ at k stage,

$$\tilde{H}_k^{\text{reg}} = \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} + E_k \succ 0, \quad k = 0, 1, \dots, N-1, \quad (1.2)$$

where E_k could be a diagonal matrix or a full matrix, depending on the regularization strategy mentioned in 1.1.1.

The full Hessian regularization is direct and usually simple to implement, but it can be too conservative without considering the definiteness of the reduced Hessian $\tilde{M}$, thus slow down the convergence.

1.1.3 Reduced Hessian regularization

According to a well-known result regarding the convexity of QPs, which is also mentioned in [4] (for proof, see [1]):

Theorem 1. *Assume that LICQ holds. The QP problem has a unique global optimum if and only if the reduced Hessian is positive definite:*

$$M = Z^\top H Z \succ 0. \quad (1.3)$$

This characterization shows that the positive definiteness of the reduced Hessian is a necessary and sufficient condition for the QP admitting a unique global optimum.

With the Theorem 1, it is reasonable to regularize the reduced Hessian $\tilde{M}$ instead of the full Hessian $\tilde{H}$, which is less conservative than the full Hessian regularization, while still effective. And from a perspective of full Hessian regularization in (1.2), the adjustment matrix is $E_k = Z_k E_k^r Z_k^\top$, where E_k^r is a positive definite matrix, added to regularize the reduced Hessian $\tilde{M}$.

However, from definition, computing the reduced Hessian directly requires a null-space basis of the equality constraints, which can be achieved via condensing in optimal control problems [7], this step generally introduces extra computational overhead.

We thus propose a Riccati-based primal regularization method, which naturally obtains the reduced Hessian $\tilde{M}$ implicitly via Riccati recursion, and regularizes the reduced Hessian $\tilde{M}$ stage-wisely during the Riccati recursion to achieve a positive definite reduced Hessian $\tilde{M}$.

1.2 Riccati-based Primal Regularization

In this section, we will introduce the idea and concept of Riccati-based primal regularization, a stage-wise regularization method for the reduced Hessian $\tilde{M}$ based on the Riccati recursion in two sense: first, Riccati recursion is used for detecting the indefiniteness of the reduced Hessian $\tilde{M}$, second, Riccati recursion

is used for regularizing the reduced Hessian $\tilde{M}$ stage-wisely to ensure a positive definite reduced Hessian $\tilde{M}$.

1.2.1 Block Reduced Hessian From Riccati Recursion

As mentioned in [8], the Riccati recursion can be interpreted as a block factorization of the KKT system in (??). In this subsection, we will further show that the Riccati recursion can be regarded as the block $\text{LDL}^\top$ factorization of reduced Hessian $\tilde{M}^i$ in (??), which justifies the stage regularization on block reduced Hessian.

From the perspective of the Riccati recursion, the backward pass can be concluded as follows:

Algorithm 1 Backward Pass of Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 
5:    $K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$ 
6:    $k_k \leftarrow -\tilde{M}_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 
7:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$ 
8:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
9: end for

```

We can find that $\tilde{M}_k = \tilde{R}_k + B_k^\top P_{k+1} B_k$ is exactly the block reduced Hessian obtained by Riccati recursion. In practice the inverse of $\tilde{M}_k$ is not computed directly; instead, the Cholesky factorization of $\tilde{M}_k$ is computed for efficiency.

From the perspective of the reduced Hessian $\tilde{M}^i$ in (??), eliminating the states through the dynamics leaves a dense reduced Hessian $\tilde{M}^i$ in the inputs $u_0, \dots, u_{N-1}$, whose density stems from the coupling of any two stages via the state transition $\Phi_{i,j}$, defined as:

$$\Phi_{i,j} = \begin{cases} \prod_{k=j}^{i-1} A_k = A_{i-1} A_{i-2} \cdots A_j, & \text{if } i > j, \\ I, & \text{if } i = j. \end{cases}$$

The backward pass in Algorithm 1 is exactly a block $\text{LDL}^\top$ factorization of this $\tilde{M}^i$, carried out without ever forming it explicitly. The connection is made

through the Schur complement: each cost-to-go matrix P_k is the Schur complement obtained by eliminating the states of the later stages, so that the pivot block $\tilde{M}_k$ is precisely the Schur complement of stage k in $\tilde{M}^i$ after all later stages have been eliminated. Consequently, the recursion $P_k = \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$ is nothing but the successive formation of these Schur complements.

Collecting the pivots $\tilde{M}_{N-1}, \dots, \tilde{M}_0$ into the block-diagonal factor D and eliminating the stages in the same order as the backward pass, from stage $N-1$ down to stage 0, the block LDL[⊤] factorization of the reduced Hessian $\tilde{M}^i$ can be expressed as:

$$\tilde{M}^i = L \begin{bmatrix} \tilde{M}_{N-1} & 0 & \cdots & 0 \\ 0 & \tilde{M}_{N-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{M}_0 \end{bmatrix} L^\top$$

where the accumulated back-substitution coefficients, built from the feedback gains K_k and the transitions $\Phi_{i,j}$, form the unit lower-triangular factor L :

$$L = \begin{bmatrix} I & 0 & \cdots & 0 & 0 \\ -B_{N-2}^\top K_{N-1}^\top & I & \cdots & 0 & 0 \\ -B_{N-3}^\top \Phi_{N-1,N-2}^\top K_{N-1}^\top & -B_{N-3}^\top K_{N-2}^\top & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -B_0^\top \Phi_{N-1,1}^\top K_{N-1}^\top & -B_0^\top \Phi_{N-2,1}^\top K_{N-2}^\top & \cdots & -B_0^\top K_1^\top & I \end{bmatrix}.$$

This establishes the connection between positive definiteness of the reduced Hessian $\tilde{M}^i$ and the positive definiteness of the block reduced Hessians $\tilde{M}_k$ obtained from Riccati recursion, which justifies the stage regularization on block reduced Hessian $\tilde{M}_k$.

1.2.2 Stage Regularization in Riccati Recursion

With this connection, we can regularize the reduced Hessian $\tilde{M}^i$ without explicitly computing it, but by stage-wisely applying the regularization to the block reduced Hessian $\tilde{M}_k^i$ during the Riccati recursion,

As mentioned in 1.2.1, the Cholesky factorization of $\tilde{M}_k^i$ is computed for efficiency, which requires the positive definiteness of $\tilde{M}_k^i$. This can be regarded as a detection of the indefiniteness of the reduced Hessian $\tilde{M}^i$. And when the Cholesky

factorization fails, which indicates that the reduced Hessian $\tilde{M}^i$ is indefinite, we can regularize the reduced Hessian block $\tilde{M}_k^i$ by adding a block adjust matrix E_k^i , i.e. $\tilde{M}_k^i + E_k^i$. The E_k^i can be computed from the methods mentioned in 1.1.2. This gives a modification of the backward pass of Riccati recursion and can be concluded as follows:

Algorithm 2 Backward Pass of Modified Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ ,  $\lambda > 0$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
5:   if factorization fails then
6:      $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$  ▷ Apply regularization
7:      $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
8:   end if
9:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 

10:   $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$ 
11:   $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 

12:   $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$ 
13:   $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
14: end for

```

Since the block reduced Hessians $\tilde{M}_k$ are exactly the pivots of the block $L^\top DL$ factorization of $\tilde{M}^i$, enforcing the positive definiteness of every $\tilde{M}_k$ during the backward pass is equivalent to regularizing the reduced Hessian $\tilde{M}^i$ itself, and thus recovers the SOSC in Theorem 1. The regularization is therefore applied only where it is needed while the structure-exploiting complexity of the Riccati recursion is preserved.

1.3 Improved Stage Regularization Strategy

In this section, we will analyze the case where the common-used minimum perturbation strategy in regularization fails in stage regularization, which causes a severe condition number issue in the preceding stages due to the propagation of the negative curvature through the stages. This motivates us to propose an improved stage regularization strategy that avoids minimum perturbation and

takes the propagation into account.

1.3.1 Failure of Minimum-Perturbation Regularization

In this subsection, we will show that the minimum perturbation strategy, i.e. eigenvalue modification with the minimum Frobenius norm of the adjustment matrix E_k [1], which is commonly the first choice for regularization with the least modification of the original problem, can cause condition number issue in the stage regularization by introducing a general example.

Suppose at the stage k , the reduced Hessian block $\tilde{M}_k$ is not positive definite. Without loss of generality, we can assume $\tilde{M}_k$ has only one negative eigenvalue λ_j , and the corresponding eigenvector is v_j . The minimum perturbation strategy is applied as:

$$E_k = (-\lambda_j + \epsilon)v_j v_j^\top, \quad (1.4)$$

where the ϵ is a small positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By Sherman-Morrison formula, the inverse of the regularized reduced Hessian block can be expressed as:

$$\begin{aligned} (\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\lambda_j - \epsilon}{\epsilon \lambda_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) v_j v_j^\top, \end{aligned}$$

and the update of cost-to-go matrix P_k follows as:

$$\begin{aligned} P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k, \end{aligned} \quad (1.5)$$

note that this negative rank-1 modification can already render P_k itself indefinite, independently of its downstream effect on $\tilde{M}_{k-1}$, whenever the perturbation magnitude exceeds the smallest eigenvalue of P_k^{unreg} . This can happen even when the perturbation on $\tilde{M}_k$ is moderate, provided that the negative-curvature direction v_j is strongly coupled to the state through Γ_k .

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage as:

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) z_j z_j^\top, \quad (1.6)$$

where $z_j = B_{k-1}^\top \Gamma_k^\top v_j$ is a column vector.

And the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as:

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\lambda_j|)}{\epsilon|\lambda_j|}, \quad (1.7)$$

where $c = \frac{\|z_j\|^2}{\lambda_{\min}(\tilde{M}_{k-1}^{\text{unreg}})}$ as an $O(1)$ constant.

Notably, as the ϵ is a small positive value, the presence of the negative rank-1 update $-\frac{1}{\epsilon} z_j z_j^\top$ will dominate, not only severely deteriorates its condition number, but also likely to make $\tilde{M}_{k-1}$ indefinite, which can further cause a domino effect through the stages.

1.3.2 Insights and Proposed Methods

This issue reveals two insights in stage regularization:

- The minimum modification objective [9] is not the foremost consideration for stage regularization, as it can cause severe condition number issue in the preceding stages. A trade-off between perturbation norm and condition number must be carefully treated.
- The cost-to-go matrix P_k must be incorporated into the regularization design, as the severity of propagation depends on the coupling structure between stages, which a purely local scheme cannot capture.

From the first insight, instead of the minimum perturbation strategy, we can take the absolute value of the negative eigenvalues or inertia as the perturbation magnitude, which can avoid the severe condition number issue in the preceding stages. The technic is also mentioned in [10] as MIRROR for eigenvalue modification, and [9] as Type-II for modified Cholesky factorization. From the second insight, we can incorporate the cost-to-go matrix P_k into the strategy design, which also can be achieved by adding an adjustment matrix $E_{k+1,P}^i$ to the cost-to-go matrix P_k directly, so that the propagation of the negative curvature can be taken into account in the regularization design.

The resulting improved stage regularization strategy can be summarized as

follows:

Algorithm 3 Backward Pass of Improved Modified Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ ,  $\lambda > 0$ 

2: for  $k = N - 1$  down to 0 do
3:    $P_{k+1}^{\text{reg}} \leftarrow P_{k+1}$  ▷ Default with no regularization
4:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\text{reg}} B_k$ 
5:    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
6:   if factorization fails then
7:      $E_{k+1,P} \leftarrow \mathcal{E}(P_{k+1})_\lambda - P_{k+1}$  ▷  $\mathcal{E}(\cdot)$ : MIRROR or Type-II strategy
8:      $P_{k+1}^{\text{reg}} \leftarrow P_{k+1} + E_{k+1,P}$  ▷ Regularize cost-to-go matrix
9:      $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\text{reg}} B_k$ 
10:     $E_{k,M} \leftarrow \mathcal{E}(\tilde{M}_k)_\lambda - \tilde{M}_k$ 
11:     $\tilde{M}_k \leftarrow \tilde{M}_k + E_{k,M}$  ▷ Regularize reduced Hessian block
12:     $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
13:   end if
14:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1}^{\text{reg}} A_k$ 

15:    $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$ 
16:    $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1}^{\text{reg}} r_{\text{eq}}^k + p_{k+1}) \right)$ 

17:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1}^{\text{reg}} A_k + \Gamma_k^\top K_k$ 
18:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1}^{\text{reg}} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
19: end for

```

In Algorithm 3, when the Cholesky factorization fails, the regularization acts at two levels. The term $E_{k+1,P}$ proactively convexifies the cost-to-go matrix that enters the current stage, so that the negative-curvature propagation identified in the second insight is accounted for in $\tilde{M}_k$, Γ_k , and P_k consistently. In contrast, $E_{k,M}$ acts as a local safeguard on the reduced Hessian block and is to ensure that the Cholesky factorization succeeds. Both perturbations are constructed from the absolute values of the negative eigenvalues (the mirror strategy) rather than the minimum-modification rule, which keeps the condition number of the regularized blocks bounded, as motivated by the first insight. When no factorization failure occurs, the recursion reduces to the standard Riccati backward pass.

1.4 Asymptotic Inactivity of the Regularization

In this section, we will prove that the reduced Hessian regularization via Riccati recursion would be asymptotically inactive, which means that the regularization will be inactive when the iterates are sufficiently close to the optimal solution (y^*, λ^*) , following a similar proof strategy to [11].

We will start with the regularized KKT formulation where the null space method is applied before the Schur complement to the (??), the resulting formulation is given as:

$$\begin{bmatrix} W & Z^\top G^\top \\ GZ & -\Lambda_i^{-1} T_i \end{bmatrix} \begin{bmatrix} \Delta y_z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} r_{\text{stat, redu}}^i \\ \tilde{r}_{\text{ineq, redu}}^i \end{bmatrix}, \quad (1.8)$$

where

$$\begin{aligned} W &= M + E^i, \\ r_{\text{stat, redu}}^i &= Z^\top r_{\text{stat}}^i - Z^\top H Y \Delta y_y, \\ \tilde{r}_{\text{ineq, redu}}^i &= r_{\text{ineq}}^i - G Y \Delta y_y - \Lambda_i^{-1} r_{\text{comp}}^i, \end{aligned}$$

this formulation eliminates the equality constraints, recasting the optimal control structure into a general QP form, and yields a saddle-point system coupling the reduced primal variable Δy_z (with regularized reduced Hessian W) and the dual variable $\Delta \lambda$.

We will make the following assumptions for the proof:

Assumption 1. (y^*, λ^*) is a local (or global) minimizer for (??).

Assumption 2. Each local minimizer (y^*, λ^*) for (??) is isolated.

Assumption 3. Linear independence constraint qualification (LICQ) holds at the minimizer (y^*, λ^*) .

Assumption 4. Strict complementarity holds at the minimizer (y^*, λ^*) .

The Assumptions 1-3 imply the second order sufficient condition (SOSC) for local optimality, so we have the following lemma 2:

Lemma 2. Under Assumptions 1-3, for all $v \in C(y^*, \lambda^*) \setminus \{0\}$, it holds that $v^\top H v > 0$.

Proof. With Assumption 1, The second order necessary condition (SONC) for local optimality holds, which states that for all $v \in C(y^*, \lambda^*)$, we have $v^\top H v \geq 0$.

We proceed by contradiction. Suppose there exists $v \in C(y^*, \lambda^*) \setminus \{0\}$ such that $v^\top H v = 0$. Consider the point $y(\alpha) = y^* + \alpha v$ for $\alpha > 0$. Because the constraints are linear and v is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α .

Since the objective function of ?? is quadratic. Expanding $f(y(\alpha))$ yields:

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top v + \frac{1}{2} \alpha^2 v^\top H v$$

As $\nabla f(y^*)^\top v = 0$, given the assumption that $v^\top H v = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $v^\top H v > 0$ must hold. $\square$

Proposition 3. *Under Assumption 1-4, given the sequence $\{W^i\}$ constructed by the algorithm, as $\{(y^i, \lambda^i)\} \rightarrow (y^*, \lambda^*)$, $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large.*

Proof. Under Assumption 4, the critical cone $C(y^*, \lambda^*)$ reduces to the tangent space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{v \mid Fv = 0, G_{\mathcal{A}}v = 0\}$.

For any $v \in \{v \mid \|v\| = 1\}$, we have the corresponding quadratic form of the reduced Hessian as:

$$\begin{aligned} v^\top \tilde{M}^i v &= d^\top (H + G^\top T_i^{-1} \Lambda_i G) d \\ &= d^\top H d + \sum_j \frac{\lambda_j}{s_j} (G_j d)^2 \\ &= d^\top H d + \sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \end{aligned}$$

where $d = Zv \in \{d \mid Ad = 0\} \setminus \{0\}$.

We now proceed by contradiction, Assume $\exists v_i \in \{v \mid \|v\| = 1\}, v_i \rightarrow \bar{v} : v_i^\top \tilde{M}^i v_i \leq 0$. We can distinguish two cases:

For $G_{\mathcal{A}} \bar{d} = 0$, this indicates $d_i \in C(y^*, \lambda^*) \setminus \{0\}$. $d_i^\top H d_i > 0$ follows by Lemma 2. Rest of the terms are non-negative, so we have $\bar{v}^\top \tilde{M}^i \bar{v} > 0$, contradict with assumption.

For $G_{\mathcal{A}} \bar{d} \neq 0$, the term $\sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \rightarrow \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}^i v_i \rightarrow \infty > 0$, contradict with assumption.

Therefore, we can conclude that $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large, i.e., $E_i = 0$. $\square$

Proposition 3 shows the regularization is asymptotically inactive: once the iterates are close to (y^*, λ^*) , $E^i = 0$ and the regularized recursion coincides with the unregularized one. The solver thus recovers the exact Newton step and inherits the local convergence rate of the underlying interior-point method, without the regularization affecting its asymptotic behavior.

Chapter 2

Numerical Experiments

In this chapter, we would focus on the experiments and corresponding results regarding the proposed regularization methods. We first introduce the experiment metrics, including iteration profile and flop unit estimation in Section 2.1, then we present the experiment setup, including the prototype solver implementation and general setup of the experiments in Section 2.3. To conclude, we present the numerical comparison of the proposed regularization methods with other existing methods in the literature.

2.1 Experiment Metrics

In this section, we will introduce the metrics used to evaluate the performance of different regularization strategies in our experiments. Since we care about both the convergence behavior and the factorization cost caused by different regularization methods, we will use two complementary metrics: iteration profile and flop unit estimation. The former one, iteration profile, is used to evaluate how many iterations each method takes to converge on a set of test instances, while the latter one, flop unit estimation, is approximated by the number of modified Cholesky factorization of the reduced Hessian, which is used to evaluate the factorization cost of each method.

2.1.1 Iteration Profile

Let $\mathcal{P}$ denote the set of test instances and $\mathcal{M}$ the set of candidate regularization strategies. For a method $m \in \mathcal{M}$ and an instance $p \in \mathcal{P}$, let $i_{m,p}$ be the number of outer iterations at which m satisfies the convergence criterion on p . A run that reaches the iteration cap $i_{\max}$ without converging, or that stalls early, is treated

as unsolved and we set $i_{m,p} = \infty$.

The *iteration profile* of method m is

$$\pi_m(k) = \left| \{ p \in \mathcal{P} : i_{m,p} \leq k \} \right|, \quad k = 1, \dots, i_{\max}, \quad (2.1)$$

i.e. the number of instances that m solves within a budget of k iterations. Equivalently, $\pi_m(k)/|\mathcal{P}|$ is the empirical cumulative distribution of the solve-iteration count over solved instances. By construction π_m is non-decreasing in k , and its value at the cap, $\pi_m(i_{\max})$, equals the total number of instances solved by m under the budget.

The profile is read along two axes. The rate at which $\pi_m(k)$ rises with k reflects the *efficiency* of method m : a curve that climbs steeply at small k solves most instances with few iterations. The height of the plateau, $\pi_m(i_{\max})$, reflects the *robustness* of the method: it is the fraction of the test set solved within the prescribed budget. Because every method is evaluated under the same cap $i_{\max}$ and the same convergence criterion, the curves are directly comparable across $\mathcal{M}$.

This metric gives a summary of how reliably and how quickly each strategy converges across the test set.

2.1.2 Flop Unit Estimation

Iteration counts alone do not reflect the work performed per iteration, which differs substantially across regularization strategies: an eigenvalue-based correction and a modified-Cholesky correction may reach convergence in a comparable number of iterations while incurring very different arithmetic cost. To compare methods on a common scale, we express all dense linear-algebra work in a single flop unit.

We take one modified Cholesky factorization of the block reduced Hessian as the unit. For a control dimension n_u , factorizing an $n_u \times n_u$ matrix costs approximately $\frac{1}{3}n_u^3$ flops, hence

$$\text{unit} \approx \frac{1}{3} n_u^3. \quad (2.2)$$

The modified factorization differs from the plain one only by lower-order diagonal updates, so to leading order it also costs one unit.

The remaining primitives are expressed relative to this unit. A symmetric

eigenvalue decomposition of the same block costs about $9n_u^3$ flops [10], i.e.

$$\frac{9n_u^3}{\frac{1}{3}n_u^3} = 27 \text{ units.} \quad (2.3)$$

With this unit defined, the arithmetic cost of a solve, including the regularization applied at each iteration, can be expressed on a common scale and accumulated over the iteration profile.

2.2 Prototype Solver Implementation

In order to evaluate the proposed regularization methods in the context of OCP QPs, we aim at achieve a similar iteration behavior as HPIPM in convex QPs, which is the state-of-the-art Riccati-based IPM solver for OCP QPs.

As the basic infrastructure for experiment, a Riccati-based IPM solver in Python with conditional Mehrotra’s predictor-corrector as barrier update strategy and FTB as step size control strategy is implemented.

And in FTB, a heuristic is used to determine the primal step size α^{prim} and dual step size α^{dual} for updating variables instead of using a fixed fraction parameter η , which can be expressed as follows:

$$\alpha = \begin{cases} \alpha_{\max} \left[(1 - \alpha_{\max}) \kappa_{\eta} + \alpha_{\max} \hat{\eta} \right], & \alpha_{\max} < 1, \\ \alpha_{\max}, & \alpha_{\max} = 1, \end{cases} \quad (2.4)$$

where $\kappa_{\eta} = 0.99$, $\hat{\eta} = 1 - 10^{-7}$, and $\alpha_{\max} \in (0, 1]$ denotes the largest step length. The bracketed factor is a combination of the conservative constant κ_{η} and the near-unit value $\hat{\eta}$, weighted by $\alpha_{\max}$.

For a fair comparison, the corresponding solver settings in HPIPM are also considered. We define the settings used in the prototype solver, referred to as **CUSTOMIZED** mode in HPIPM, the details are listed in Table 2.1.

Table 2.1: Detailed settings for defined CUSTOMIZED Mode.

Parameter	Value	Parameter	Value
mu0	1e1	lam_min	1e-12
alpha_min	1e-12	t_min	1e-12
res_g_max	1e-6	tau_min	1e-12
res_b_max	1e-8	lam0_min	1e-12
res_d_max	1e-8	t0_min	1e-12
res_m_max	1e-6	warm_start	0
dual_gap_max	1e15	abs_form	0
iter_max	30	comp_dual_sol_eq	1
stat_max	30	comp_res_exit	1
pred_corr	1	comp_res_pred	1
cond_pred_corr	1	split_step	1
itref_pred_max	0	m_safe	0.5
itref_corr_max	0	var_init_scheme	0
reg_prim	0.0	t_lam_min	2
square_root_alg	0	t0_init	2
lq_fact	0	update_fact_exit	1

As one can notice, the CUSTOMIZED mode largely coincides with the SPEED mode of the original HPIPM, differing in two respects. First, the square-root algorithm is disabled (`square_root_alg=0`), so that the Riccati recursion is carried out directly rather than on Cholesky factors. Second, the primal and dual step sizes are computed separately (`split_step=1`) instead of sharing a single step length. In both modes no iterative refinement is applied to the predictor or corrector steps (`itref_pred_max=itref_corr_max=0`) and no default primal regularization is added (`reg_prim=0`), while Mehrotra’s predictor–corrector is retained and applied conditionally (`pred_corr=cond_pred_corr=1`).

To verify this similarity, we collected 24 convex QP subproblems (three from each of the 8 OCPs in the benchmark) and compared the iteration behavior of the CUSTOMIZED-mode HPIPM against that of the prototype solver. The resulting iteration profiles are shown in Figure 2.1: the two profiles coincide exactly on the convex QPs, confirming that on convex QPs the prototype solver reproduces the iteration behavior of HPIPM.

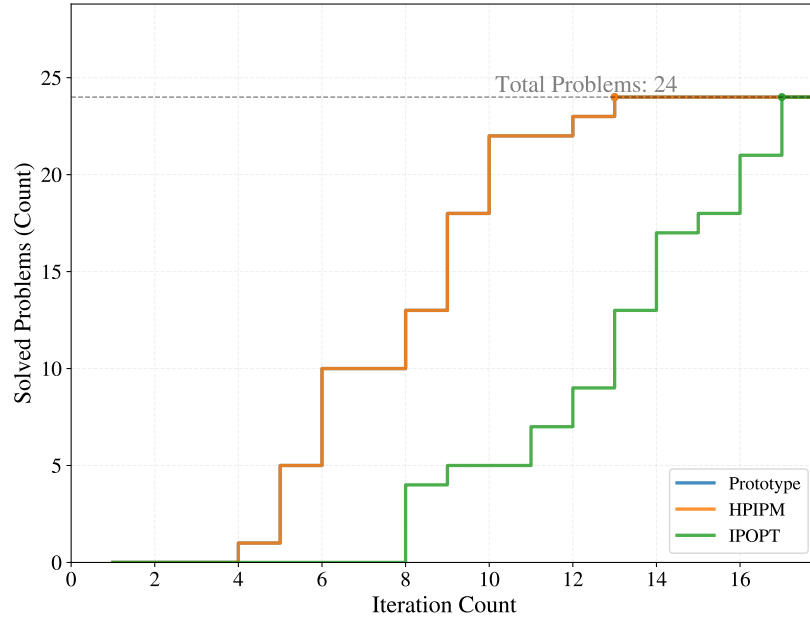


Figure 2.1: Iteration profiles on 24 convex QPs. The prototype solver is compared with HPIPM, showing identical iteration behavior.

2.3 Experiment Setup

In this section, the experiment setup is discussed. We will introduce the general setup of the experiments, including the regularization methods considered in the solver and the convergence criterion for experiments.

2.3.1 Experimented Regularizations

In the experiment, we will first classify the regularization methods to organize the following comparison. According to the Section 1.1, The categories of regularizations can distinguished by where the regularization strategies are applied, as Full Hessian Regularization, Riccati-based Regularization and Inertia correction.

2.3.2 General setup

We set the maximal iteration $i_{\max}$ as 100. For convergence criterion, we have $1e-6$ for stationary residual r_{stat} and complementarity residual r_{comp} , $1e-8$ for equality residual r_{eq} and inequality residual r_{ineq} .

Furthermore, as mentioned in Section ??, more careful step-size control is needed in nonconvex programs to aid convergence. We therefore add a funnel

line search to the prototype solver for step-size control. The algorithm is given in Algorithm ??, and the corresponding parameters are listed in Table 2.2:

Table 2.2: Parameters of the funnel line search.

Symbol	Value	Role
β	0.5	Backtracking factor, $\alpha_p \leftarrow \beta \alpha_p$
$n_{\max}$	10	Maximum backtracking iterations
$\theta_{\max}^0$	100	Initial max funnel bound
κ_s	10^{-1}	Switching constant, $\alpha_p(-\rho) > \kappa_s \theta^2$
γ	10^{-4}	Armijo constant for f-step, $f^+ \leq f + \gamma \alpha_p \rho$
κ_θ	0.999	h-step acceptance factor, $\theta^+ \leq \kappa_\theta \theta_{\max}$
κ_u	0.5	Funnel-update weight, $(1 - \kappa_u)\theta_{\max} + \kappa_u \theta^+$

2.4 Numerical Comparison of Primal Regularizations

In this section, we present the numerical results of the proposed regularization methods on nonconvex QPs with different solvers. We default Mehrotra’s probing as the barrier update strategy, the proposed funnel line search as the step size control strategy, and compare the iteration profile and flop unit consumption of different regularization methods.

2.4.1 Comparison in Riccati-based regularizations

2.4.2 Comparison with Full Hessian Regularizations

2.4.3 Comparison with Inertia Correction

2.5 Possible Discussion

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